

A Deterministic Symbolic Sequence from Phonetic Letter Encoding: Sublinear Complexity and an LCR near π

Bahaa Budargham

*Engineering of Alphabets (EOA), a research program
of Neurolabs Inc., Montreal, Canada*

ORCID: <https://orcid.org/0009-0005-2955-3703>

Email: bdarghamneurolabs@gmail.com

Abstract

We report a deterministic symbolic sequence generated from a single English letter name using a recursive operator on phonetic encodings. The first 2,000 symbols exhibit sublinear factor complexity, zero topological entropy, near-deterministic letter transitions, and a limiting growth ratio close to π . We provide the sequence, its statistics, and open questions. We invite collaboration to identify the underlying substitution or automaton and to explain the observed phenomena.

1. Introduction

The Engineering of Alphabets (EOA) program studies a deterministic recursive operator that maps spoken letter names into numerical sequences. Under a specific phonetic encoding, the operator produces a symbolic string over a subset of the English alphabet. This note reports the symbolic dynamics of that string for one letter name.

The sequence is not random. It has a small alphabet, highly structured growth, and a limiting ratio that is numerically close to π . The 24th term of the sequence contains approximately 1,116,231,938,318 letters (over one trillion). We release the first 2,000 letters for independent analysis.

We do not claim to understand why the sequence has these properties. Our goal is to present the data and invite experts in symbolic dynamics, combinatorics on words, and formal languages to help identify the underlying structure.

2. The Sequence and Generation

The sequence is generated by a fixed deterministic operator applied to the spoken name of a single English letter. The operator itself is withheld under staged disclosure (see Section 6). The encoding maps each letter name to a short phonetic string; the operator then iterates a recursive transformation that produces the symbolic output.

The released prefix uses 16 distinct letters:

a, b, c, d, e, g, h, i, k, l, o, s, t, u, w, y

The remaining 10 letters of the English alphabet do not appear in this prefix. The first 500 symbols are:

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gzhlvzggjzyihjgheqvzgzhlvzhlbjzgyhzizygjzjbzhlvgjzjehjdzzgheqvzgzhlvzggjzyihjghe  
qvzggjzyihjgheqvypzajukbjzgzhlvyhzgjjzgyzgyhzzhlvbjzgygpzajukbjzggjzyihjgheqv  
zhlvbjzgehjjzbdzgzgzhlvvgjzjehjdzzgheqvzgzhlvzggjzyihjgheqvzgzhlvzhlvbjzgyhzizy  
gjzjbzhlvgjzjehjdzzgheqvzgzhlvzhlvbjzgyhzizygjzjbzhlvgjzjehjdzzgheqvzhzzhlpzzgzaa  
bjuvzbygpzajukbjzgzhlvzggjzyihjgheqvzhzgjzgzgzhlvbjzgzgzhlvizyzyghzzgzhlvhgzgjjz
```

The full 2,000-letter sequence, code, checksums, and generation notes are available at Zenodo.

Term lengths for the first 12 terms are:

Term n	New letters	Cumulative letters
1	1	1
2	3	4
3	8	12
4	26	38
5	86	124
6	276	400
7	869	1,269
8	2,719	3,988
9	8,506	12,494
10	26,644	39,138
11	83,563	122,701
12	262,283	384,984

The growth ratio L_n / L_{n-1} converges to approximately 3.173.

3. Symbolic Dynamics Observations

We analyzed the first 2,000 symbols as a finite word over the 16-letter alphabet.

Factor complexity

The number of distinct subwords of length k is:

k	$p(k)$
1	16
2	57
3	110
4	157
5	203

For a random sequence over 16 letters, $p(2)$ would be near 256 and $p(3)$ near 4,096. The observed values are far smaller and grow slowly. The growth is sublinear, indicating zero topological entropy. This is the signature of a morphic or automatic sequence.

Repeated factors

A greedy parsing covers 99.9% of the word using only 65 recurrent factors. The most frequent are:

- zgzhlv (50 occurrences)
- gheqv (50 occurrences)
- zggjzyihgheqv (32 occurrences)

The longest repeated factor has length 79. This long-range repetition is a strong self-similarity signal.

Transition matrix

For many letter pairs, the succession is nearly deterministic:

- $l \rightarrow v$ (93/93)
- $q \rightarrow v$ (50/50)
- $p \rightarrow z$ (35/35)
- $d \rightarrow z$ (27/27)

Only a few letters (z, g, j, y) show genuine branching. This profile is typical of a morphic word with a small branching core.

Return times

The dominant motif zgzhlv reappears 50 times with irregular gaps (mean 40, standard deviation 22, range 6–88). This bounded but irregular return structure is consistent with a primitive substitution.

4. The Growth Rate

The term lengths grow by a factor that converges to approximately 3.173. If the sequence is generated by a finite substitution, this growth rate is the Perron eigenvalue of the substitution matrix. The value is numerically close to π (3.14159) to five decimal places, but we do not claim exact equality.

We do not know whether this eigenvalue is algebraic, transcendental, or a numerical coincidence. This is one of the open questions we hope to answer with collaborators.

5. Open Questions

We invite experts to investigate the following:

- Is the sequence morphic? If so, what are the substitution rules σ ?
- Can the Perron eigenvalue be computed exactly? Is it algebraic? Does it have a closed form?
- Why is it close to π ? Is this a deep Diophantine approximation or a coincidence?
- What is the finite automaton that generates the sequence?
- What is the relationship to the 13-dimensional basis found in the generating functions of the underlying numerical sequences?
- What are the return words and their frequencies? Can they be derived from a finite grammar?
- Does the sequence have a known classification in symbolic dynamics (e.g., Sturmian, quasi-Sturmian, S-adic, automatic)?

6. Collaboration

We are open to collaboration under a standard academic agreement that respects the staged disclosure of the full operator. We can provide:

- Additional terms up to the 24th (over one trillion letters) for statistical analysis.
- Access to the operator for qualified researchers under the access terms described in Part I of the EOA program.
- Co-authorship on papers that produce correct, reproducible theoretical explanations of the phenomena described here.
- Independent publication rights with acknowledgement for researchers who prefer to work alone.

We are particularly interested in collaboration with symbolic dynamicists, combinatorists on words, and researchers in formal languages and automata theory.

Contact: [Bahaa Budargham](#),

7. Data Availability

The full 2,000-letter sequence, the generation code, checksums, and additional plots are available at:

- GitHub: <https://github.com/bahaa-budargham/ea-symbolic-sequences>

The data is released under <https://doi.org/10.5281/zenodo.22882603>. The operator itself is withheld under staged disclosure; the sequence is released for independent analysis.

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